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SCIENCE

COMPSCI 713: AI Fundamentals
Introduction

Instructor:
Dr Thomas Lacombe
Dr. Xinyu Zhang

What Is Artificial Intelligence?

Artificial intelligence (AI) is the simulation of human intelligence in machines that are programmed to think and act like humans.

Learning, reasoning, problem-solving, perception, and language comprehension are all examples of cognitive abilities.

Artificial Intelligence is a method of making a computer, a computer-controlled robot, or a software "think" intelligently like the human mind.

AI is accomplished by studying the patterns of the human brain and by analyzing the cognitive process. The outcome of these studies develops intelligent software and systems.

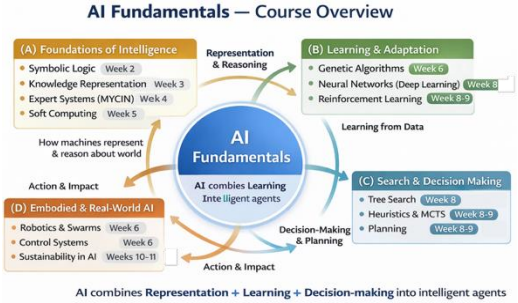
Weak AI vs. Strong AI

Weak AI (Narrow AI)
Weak AI refers to AI systems that are designed to perform specific tasks and are limited to those tasks only. These AI systems excel at their designated functions but lack general intelligence. Examples of weak AI include voice assistants like Siri or Alexa, recommendation algorithms, and image recognition systems. Weak AI operates within predefined boundaries and cannot generalize beyond their specialized domain.

Strong AI (General AI)
Strong AI, also known as general AI, refers to AI systems that possess human-level intelligence or even surpass human intelligence across a wide range of tasks. Strong AI would be capable of understanding, reasoning, learning, and applying knowledge to solve complex problems in a manner similar to human cognition. However, the development of strong AI is still largely theoretical and has not been achieved to date.

Timeline of the Course

Theme	Content	Importance
Foundations of Intelligence	Logic (W2) → Knowledge Representation (W3) → Expert Systems (W4) → Soft Computing (W5)	How machines **represent and reason about the world**
Learning & Adaptation	Genetic Algorithms (W6) → Deep Learning (W7) → Reinforcement Learning (W9)	How machines **learn from data and interaction**
Search & Decision Making	Tree Search, Heuristics, etc. (W8) → MCTS (W9) → Planning (W8,9)	How machines **choose optimal actions**
Embodied & Real-World AI	Robotics, Control Systems (W6) → Sustainability (W10,11)	How AI **acts in physical and societal environments**



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Introduction – COMPSCI 713

Examines the core concepts and techniques in AI, including breakthroughs in symbolic AI, machine learning, and neural networks.

This will include an introduction to pioneering breakthroughs in symbolic AI, including heuristic search, constraint satisfaction, and knowledge representation, Bayesian inference, statistical methods of machine learning, and connectionist ideas about brain-inspired neural networks, which have given rise to the contemporary developments of deep learning.

Real-world applications are presented, with a focus on AI research in Aotearoa/NZ and ethical considerations.

Teaching Team

Lecturers:

Dr Thomas Lacombe, thomas.lacombe@auckland.ac.nz
Dr Xinyu Zhang, xinyu.zhang@auckland.ac.nz

Tutors:

Zhongsheng Wang, zhongsheng.wang@auckland.ac.nz
Yuzhuo Li, yuzhuo.li@auckland.ac.nz
Zhenyu Yin, zyin418@aucklanduni.ac.nz

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Thomas's background

PhD in E.Eng. in 2018, at USMB/CUGA, France

- Industrial computer vision and ML for automatic visual quality control
- Feature engineering, feature selection

Postdoc at UoA in 2019-2020

- Automated hyper-parameter tuning for machine learning with data streams

Since 2021

- Teaching fellow in the School of Computer Science
- Interest in interactions between Generative AI and education

UNIVERSITÉ SAVOIE MONT BLANC

Communauté UNIVERSITÉ Grenoble Alpes

https://scikit-ika.github.io/

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Xinyu Zhang's background

PhD in 2021, at Tongji University, China

- Machine Learning and Computer Vision
- Un/Semi-supervised learning; Domain adaptation

Senior Research Scientist in 2021-2024, Baidu Inc., China

- Foundation model pre-training; Human-centric AI

Research Fellow at The University of Adelaide in 2024-2025

- Generative AI; image/video generation

Since 2025

- Lecturer in the School of Computer Science, UoA
- Generative AI

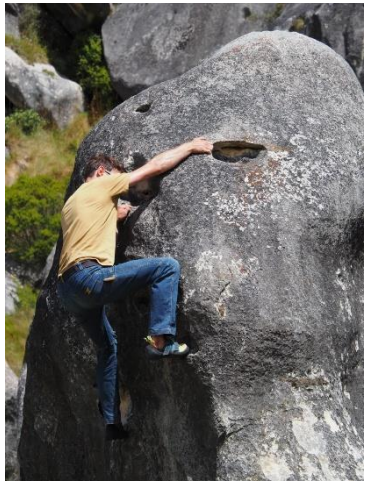
Introduce yourself

Before we start talking about the course, get to know your classmates!

Who are you?

Where are you from?

Something you like to do!



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Lectures times and locations

When	Where
Monday 12:00 to 13:00	BLT100/106-100
Thursday 14:00 to 16:00	BLT100/106-100

This course is a standard 15-point course and students are expected to spend **in average 10 hours per week** involved in each 15-point course that they are enrolled in.
In total, expect 150 hours of work over 15 weeks (12 teaching weeks and three weeks of study break and examinations), broken down into as follows: 36 hours of lectures, 60 hours of reading and thinking about the content and 54 hours of work on assignments and/or test preparation. It is important to consider that some weeks may require more hours of study for completing assignments and that tests and examinations may be scheduled outside of normal class or working hours.
You can expect a typical week of work to be broken down as follows:

- 3 hours of lecture
- 4 hours of reading and thinking about the content
- 3 to 4 hours of work on assignments/projects and/or test preparation

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Tentative Course Schedule Semester 1 2026

	Monday 12pm-1pm	Thursday 2pm-4pm
Week 1 Begins 02/03	Intro Lecture (Thomas & Xinyu)	Lecture (Xinyu) What is AI and how did it develop until now?
Week 2 Begins 09/03	Lecture (Xinyu) Symbolic logic	Lecture (Xinyu) Logic Theorist & Logical NN
Week 3 Begins 16/03	Lecture (Xinyu) Knowledge representation	Lecture (Xinyu) Knowledge representation
Week 4 Begins 23/03	Lecture (Xinyu) MYCIN (expert systems)	Lecture (Xinyu) ID3 to XGBoost (decision trees, tree induction, ensembles)
Week 5 Begins 30/03	Lecture (Xinyu) Soft Computing (Bayesian reasoning, Fuzzy logic & control)	Workshop 1: Brainstorm and propose group project
Mid-Sem Break Begins 06/04		
Mid-Sem Break Begins 13/04		
Week 6 Begins 20/04	Lecture (Xinyu) NEAT (Genetic algorithms, neuro-genetic control)	Lecture (Xinyu) Robots! (layered control, BigDog & space, swarms, robot soccer)
Week 7 Begins 27/04	ANZAC Day (Observed) No lecture	Workshop 2: Writing a literature review in the field of AI
Week 8 Begins 04/05	Lecture (Thomas) ALPHAGO (Deep Learning, RL)	Lecture (Thomas) Search Algorithms I (Tree search, Uninformed search, informed search, heuristics, A*)
Week 9 Begins 11/05	Lecture (Thomas) Search Algorithms II (adversarial search, MCTS)	Lecture (Thomas) Reinforcement Learning (MDP, RL principles, intro to DRL)
Week 10 Begins 18/05	Tentative: Guest Lecture (TBD)	Lecture (Thomas) AI and Sustainability: GraphCast (Graph data, GNNs)
Week 11 Begins 25/05	Lecture (Thomas) Sustainabilty in AI: Self-supervised learning (SSL, pretext tasks)	Lecture (Thomas) Sustainability in AI: Data Streams/C-Learning (data streams, incremental / lifelong / continual vs batch learning)
Week 12 Begins 01/06	King's Birthday (no lecture)	Paper discussion session on selected presentations (6 or 7 papers)

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Key Topics

- Symbolic Logic
- Knowledge Representation
- Rule base
- Expert System
- Decision Trees
- Soft Computing
- Genetic Algorithms
- Embodied AI
- Search algorithms
- Graph Neural Networks
- Markov Chain
- Reinforcement Learning
- Self-supervised learning
- Continual learning
- Time Series/Data Streams

Communication in the course

- **Teaching team to students**:

  Canvas Announcements
  - Check them/your UoA email daily (please make sure that your personal notification settings is set to ASAP for announcements).

- **Students to teaching team**:

  Ed Discussion Forum & Email
  - Ed Discussion forum for general course questions, e.g., content related, general admin question.
  - Email for specific/private/urgent requests/meeting request.

- **Email guidance**:
  - Email the right person: course coordinator (admin-related) vs. lecturer (content-related)
  - Use descriptive email subjects: [CS713] subject of the email
  - Be short & to the point, while providing enough background information.
  - Make sure the answer to your question is not given on the Canvas home page, class announcements, or the class discussions on Piazza before emailing.

- Please follow the school guidelines for communication and course expectations.

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Two Lane Assessment

Lane 1 consists of controlled assessments *of* learning. These tasks measure what you are expected to know **without** the assistance of GenAI after completing the course.

Lane 2 consists of uncontrolled assessments *for* learning. These tasks are designed to support your learning as you complete them. You are allowed to use GenAI, provided that your use is **responsible, ethical, and transparent**.

More in two-lane assessment framework

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Assignments

Assessment Overview		
Assessment Group	Lane	Weighting
Individual Presentation (recording)	Lane 2	10%
Group Project (lit. review + workshops)	Lane 2	23% + 2 x 6% = 35%
Online quizzes	Lane 2	5 x 1% = 5%
Mid-semester test	Lane 1	20%
Final Exam	Lane 1	30%

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Pass requirements

To pass the course, as well as obtaining at least 50% for your overall mark, you are also required to pass the combined Lane 1 assessments (mid-semester test & final exam) of the course and the combined Lane 2 assessements (online quizzes, presentation & group project) of the course.

The combined Lane 1 assessments are worth 50% of the overall course marks and consist of the mid-semester test worth 20% and the final exam worth 30%. To pass this component of the course you need to get at least 25 out of 50.

The combined Lane 2 assessements are worth 50% of the overall course marks and consist of the online quizzes worth 5% in total (1% per quiz), the individual presentation recording worth 10% and the group project report which is worth 35% (Deliverable 1 worth 6%, Deliverable 2 worth 6% and Deliverable 3 worth 23%). To pass this component of the course you need to get at least 25 out of 50.

Assignment: Individual

COMPSCI 713: Individual Research Presentation (10%)
Each student will select one reading paper from the provided reading list. The goal is to record a presentation on the assigned reading.

Where to find the reading papers? **Canvas assignment**

Please select your paper in the shared document linked in the assignment (limited slots per paper).

Register your choice before **Sunday 29th of March**. After this date, you will be assigned randomly to an available paper.

Video **due on 17th May**.

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Giving a good presentation

- Your video must be **6 minutes (+- 30 sec)** in total
- Be very selective in messages to emphasize so you have time to speak clearly.
- Consider drawing a figure or two to highlight your key points (it might suit better than a figure cut from the paper).
- You can find some help to prepare and deliver your presentation in the UoA Learning Essentials.
- Guidelines and tips on the Use of GenAI available in the assignment page. E.g., **you are not allowed to directly generate the video via an AI tool, or use an AI generated voice.**

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Assignment: Group

COMPSCI 713: Group Project Assignment (literature review) (35%)

1) Workshop assignment 1 (project proposal) (6%) Due date: 12th April
2) Workshop assignment 2 (lit. review outline) (6%) Due date: 10th May
3) Final lit. review report (23%) Due date: 10th June

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Individual: Online quizzes

COMPSCI 713: Online quizzes (5%)

- 5 online timed (60 min) MCQ quizzes (~ 1 every 2 weeks)
- 10 MCQ questions per quiz
- Aimed at helping you keep up with the content and pace of the course
- Good way to assess if you have the fundamentals of what was taught in previous 2-3 weeks.
- Released Friday morning, due Sunday night.

- **Use of GenAI**: We cannot check and we will not police the use of GenAI for the quizzes. However, **we highly recommend you to do the quizzes by yourself**! They are designed to check your understanding, not ChatGPT's one. Remember you will not have access to GenAI for the test and exam.

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Mid-semester test

COMPSCI 713 Mid-semester Test (20%)

- **Start Time and Date**: Tuesday 28th of April at 6:30pm
- **Duration**: 60 min (55min + 5min reading time)
- **Where**: On city campus. Rooms TBA.
- **Type of Test:** On paper.
- **Content covered**: All lectures up to the mid-term break (weeks 1 to 6 included).
- **What you must take with you to the test room:** A pen and your Student Id card.
- **What else can you can take with you to the test room:** Bottled water, and an A4 2-sided sheet of handwritten notes.
- Calculators are not allowed.

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Final exam

COMPSCI 713 Final Exam (30%)

- **Start Time and Date**: TBA during the semester (will take place in the exam period after week 13)
- **Duration**: 2 hours
- **Where**: On campus. You will be able find your room on SSO.
- **Type of Test:** On paper.
- **Content covered**: All lectures, except guest lectures + selected research papers (TBA).
- **What you must take with you to the exam room:** A blue or black pen (plus a spare), a pencil, and your **Student Campus Card (a $25 fine applies if you don't have it with you)**.
- **What else can you can take with you to the test room:** Bottled water, and an A4 2-sided sheet of handwritten notes.
- Calculators are not allowed.

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Class rep

The class rep act as an intermediary between students in the class and the lecturers and tutor. You can share with them any suggestions / complaints / remarks about the lectures.
Class rep can organise communication between students (e.g. create social media chats).

- It is mandatory to have one if the course has at least 15 students.
- The role of a lecturer is to facilitate the class rep selection process and not to impose a candidate to the class.

- We will run the class rep election on Ed Discussion!
- I will open a post where candidates can post their motivations. Then, we will proceed to a vote with a poll on Canvas if there are several candidates.

Do you need help?

The University has a range of ways to get help if something has happened to you or someone you know.

Te Papa Manaaki, Campus Care

A safe, confidential and free service where a case manager will connect and guide you through relevant support services. You can be referred by others or reach out directly for advice.

auckland.ac.nz/campus-care

University Health and Counselling Services (UHCS)

Enrolled patients can book an appointment with a GP, nurse or [illegible]ellor. Visit the website for more info, including on[illegible]olment.

auckland.ac.nz/studenthealth

Ratonga Hauātanga Taulra, Student Disability Services (SDS)

The team offer a range of services to make studying at the University an accessible and positive experience by tailoring support to suit each individual student.

disability.auckland.ac.nz.

Be Well

A range of general wellbeing self-help resources, to support your physical and mental health during your time at University.

auckland.ac.nz/be-well

Student Conduct Office

The primary reference point for all matters relating to student general misconduct, including your options for reporting and available support.

auckland.ac.nz/student-conduct

Complaints and incidents

This is where you can submit a complaint, provide feedback or report an incident to the University.

auckland.ac.nz/complaints-and-incidents

Alcohol and Drug Clinician

Confidential alcohol and drug advice and appointments are available for all students.

auckland.ac.nz/aod-clinician

Rainbow Support Adviser

Provides a holistic approach to student wellbeing for the rainbow community, offering 1:1 support and group opportunities for discussion.

equity.auckland.ac.nz/rainbow

Spiritual and religious support

The team can provide holistic pastoral care and wellbeing. They offer inclusive, compassionate support whether you have a faith or are exploring how spirituality and religion could be a part of your life.

auckland.ac.nz/spiritual-support

AUSA Advocacy

The team provide free, independent, professional advice if you are struggling with an academic, financial or personal issue.

ausa.org.nz/support/advocacy

HELP Centre

A private and confidential service providing specialist support for sexual abuse survivors. Search 'HELP' on the University website for information on the drop in service.

Phone: 0800 623 1700 (available 24/7)
helpauckland.org.nz

Puāwaitanga Counselling

A service run by trained counsellors and available to all students, providing support with feelings of anxiety, relationships, grief, addiction and more.

Phone: 0800 782 999 (7 days a week 9am-9pm)
Visit auckland.ac.nz and search 'Puāwaitanga'.

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Be Well Ulhai Hauora

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Your To-do list

- If not done already, get familiar with the course Canvas page, and join the Ed Discussion forum for the course.
- Start thinking about who you want to work with for the literature review project and what topic you want to investigate (more to come on Thursday).
- Pick your reading paper for the presentation! (First come first serve). If you do not pick a paper by 29th March, one would be assigned to you.
- For potential class reps: submit your expression of interest on Ed Discussion (once I create the post).

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Your feedback is valuable!

This is the second iteration of the course, so we welcome your feedback to improve it in the future ☺

Feedback should preferably be passed by:

Talking to your class rep so that they can report it to us

Filling the SET Evaluations (at the end of the semester)

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Welcome to COMPSCI 713!

We hope you enjoy the course!